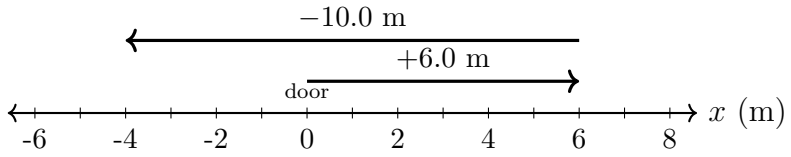
2.6 Classwork: Kinematics Review Solutions

1. Hallway walk, east positive, origin at the door.



   (a) $x = 0 + 6.0 - 10.0 = \mathbf{-4.0\ m}$, that is, 4.0 m west of the door.

   (b) Distance $= 6.0 + 10.0 = \mathbf{16.0\ m}$. Displacement $= -4.0 - 0 = \mathbf{-4.0\ m}$, or **4.0 m west**.

   Distance counts every meter walked and has no direction. Displacement is only where the student ended up, measured from the start.

   (c) Total time $= 3.0 + 5.0 = 8.0$ s.

   average speed $= \dfrac{16.0\text{ m}}{8.0\text{ s}} = \mathbf{2.0\ m/s}$ $\bar{v} = \dfrac{-4.0\text{ m}}{8.0\text{ s}} = \mathbf{-0.50\ m/s}$, or **0.50 m/s west**

   A common slip is to average the two legs' velocities, $\frac{2.0+(-2.0)}{2} = 0$*; that ignores that the legs take different times. Divide the total displacement by the total time.*

2. Usain Bolt, 100 m in 9.58 s. The race distance is exact, so the answer takes three significant figures from 9.58 s.

   (a) Step 1, the rate: average speed $= \dfrac{100\text{ m}}{9.58\text{ s}} = 10.438\ldots \approx \mathbf{10.4\ m/s}$

   Step 2, convert, carrying the unrounded value:

   $10.438\ldots\ \dfrac{\text{m}}{\text{s}} \times \dfrac{3600\text{ s}}{1\text{ h}} \times \dfrac{1\text{ km}}{1000\text{ m}} = 37.578\ldots \approx \mathbf{37.6\ km/h}$

   Also accept $\times 3.6$ *in Step 2. Rounding first,* $10.4 \times 3.6 = 37.44 \approx 37.4$ *km/h, loses the third figure — the reason to carry* $10.438\ldots$

   (b) 12.3 m/s is an **instantaneous** speed. **He starts from rest**, and the first few seconds are spent speeding up, so the average over the whole race is lower than the top speed.

   Credit any answer that names the start from rest (the acceleration phase). Also accept that he eases off slightly at the end.

3. Braking car. Knowns: $v_0 = 17.9$ m/s, $v = 0$, $a = -6.4$ m/s^2. The acceleration has two significant figures, so the answers do too.

   (a) Want t. Use $v = v_0 + at$, so $t = \dfrac{v - v_0}{a}$:

   $t = \dfrac{0 - 17.9\text{ m/s}}{-6.4\text{ m/s}^2} = 2.796\ldots \approx \mathbf{2.8\ s}$

(b) Want $x - x_0$; use $v^2 = v_0^2 + 2a(x - x_0)$:
$0 = (17.9\text{ m/s})^2 + 2(-6.4\text{ m/s}^2)(x - x_0)$, so
$x - x_0 = \dfrac{-320.41\text{ m}^2/\text{s}^2}{-12.8\text{ m/s}^2} = 25.03\ldots \approx \mathbf{25\text{ m}}$

Yes, it agrees: 25 m and 24 m differ by about 1 m, which is within the precision of a two-figure acceleration.

Also accept the average velocity route: $\bar{v} = \frac{17.9+0}{2} = 8.95$ m/s, and $8.95 \times 2.796\ldots \approx 25$ m. A negative distance means the minus sign on a was dropped in one place but not the other.

4. Dropped stone. Knowns: $v_0 = 0$, $t = 2.00$ s, $a = +9.80\text{ m/s}^2$ (down positive).

(a) Want v. Use $v = v_0 + at$:
$v = 0 + (9.80\text{ m/s}^2)(2.00\text{ s}) = \mathbf{19.6\text{ m/s}}$ downward

(b) Want $y - y_0$. Use $y = y_0 + v_0 t + \frac{1}{2}at^2$ with $v_0 = 0$:
$y - y_0 = \frac{1}{2}(9.80\text{ m/s}^2)(2.00\text{ s})^2 = 4.90 \times 4.00 = \mathbf{19.6\text{ m}}$

The two answers share the digits 19.6 only because $t = 2.00$ s; one is a speed and one a height. Also accept $\bar{v}\,t = 9.80$ m/s $\times$ 2.00 s. The usual slip is forgetting the $\frac{1}{2}$ (39.2 m) or not squaring the time (9.80 m).

5. Ball thrown up. Knowns: $v_0 = +9.80$ m/s, $a = -9.80\text{ m/s}^2$, $y_0 = 0$ at the hand (up positive).

(a) **Velocity $\mathbf{v = 0}$** for an instant. **Acceleration $\mathbf{a = -9.80\text{ m/s}^2}$**, downward, the same as everywhere else in the flight.

The common wrong answer is $a = 0$ at the top. If the acceleration were zero there, the velocity would stay zero and the ball would hang in the air.

(b) Want t when $v = 0$. Use $v = v_0 + at$, so $t = \dfrac{v - v_0}{a}$:
$t = \dfrac{0 - 9.80\text{ m/s}}{-9.80\text{ m/s}^2} = \mathbf{1.00\text{ s}}$

The ball loses 9.80 m/s every second and starts with 9.80 m/s.

(c) Want $y - y_0$. Use $\bar{v} = \dfrac{v + v_0}{2} = \dfrac{0 + 9.80}{2} = 4.90$ m/s:
$y - y_0 = \bar{v}\,t = 4.90\text{ m/s} \times 1.00\text{ s} = \mathbf{4.90\text{ m}}$

Also accept $v^2 = v_0^2 + 2a(y - y_0)$: $y - y_0 = \frac{0-(9.80)^2}{2(-9.80)} = \frac{-96.04}{-19.6} = 4.90$ m.

(d) **2.00 s.** The fall back down takes as long as the rise: $2 \times 1.00\text{ s} = 2.00$ s.

It comes back at 9.80 m/s downward, $v = 9.80 + (-9.80)(2.00) = -9.80$ m/s. Also accept setting $y = 0$ in $y = v_0 t + \frac{1}{2}at^2$: $0 = (9.80 - 4.90\,t)\,t$, so $t = 2.00$ s.